UNIT III: Cloud, Edge Computing, RFID and Blockchain in Manufacturing

Subject: 23A03703d - Smart Manufacturing

1. Introduction to Cloud Computing for Manufacturing

1.1 What is cloud computing?

Cloud computing means renting computing resources (servers, storage, databases, analytics and software) over the internet instead of owning them. The provider manages the hardware, and the user pays for what is used.

1.2 Service models

- IaaS (Infrastructure as a Service): virtual machines and storage. The user manages the operating system and applications.
- PaaS (Platform as a Service): a ready development and runtime platform, for example for building IoT applications.
- SaaS (Software as a Service): finished applications used through a browser, for example a cloud MES or ERP.

1.3 Deployment models

- Public cloud: shared infrastructure run by a provider (AWS, Microsoft Azure, Google Cloud).
- Private cloud: dedicated to one company, often in its own data centre.
- Hybrid cloud: a mix; sensitive or real-time work stays local, while analytics and archives go to the public cloud.
- Community cloud: shared by organisations with common needs, for example a regional industry group.

1.4 Benefits for manufacturing

- Scalability: add capacity for peak analysis without buying servers.
- Lower upfront cost: no large investment in data centres.

- Access from anywhere: managers and suppliers can see live data securely.
- Collaboration: plants, suppliers and customers share one data platform.
- Fast updates and access to advanced tools such as machine learning services.

1.5 Limitations

- Latency: round trips to a distant cloud can be too slow for control loops.
- Connectivity: a plant that loses its internet link may lose visibility.
- Data sovereignty and confidentiality: rules may limit where data can be stored.
- Recurring cost and vendor lock-in.

KEY: Cloud is excellent for analytics, planning and long-term storage; it is not a substitute for fast local control.

Exam Q: Give two reasons a plant might prefer a hybrid cloud model.

2. Edge and Fog Computing in Real-time Manufacturing Sy

2.1 Edge computing

Edge computing processes data close to where it is produced, on gateways, industrial PCs or devices themselves. Only useful results or summaries are sent to the cloud.

- Benefits: very low latency, works during internet outages, reduces bandwidth costs, keeps sensitive raw data on site.
- Examples: a vision system that rejects a defective bottle within 20 milliseconds; a gateway that averages 10,000 vibration samples into one value per second.

2.2 Fog computing

Fog computing is a layer between the edge and the cloud. It spreads computing across routers, local servers and gateways in the network so that data can be processed at intermediate points.

- Edge: on the machine or in the device.
- Fog: in the plant network, for example a local server that coordinates several cells.
- Cloud: centralised, for big data storage and enterprise analytics.

Layer	Location	Typical latency	Typical task
Device/edge	On or next to machine	Milliseconds	Control, filtering, immediate alarms
Fog	Plant server or network node	Tens of milliseconds	Line coordination, local dashboards
Cloud	Remote data centre	Hundreds of milliseconds or more	Analytics, ML training, enterprise reports

2.3 Real-time requirements

- Hard real-time: missing a deadline is unacceptable (for example, a safety stop).
- Soft real-time: occasional delay is tolerable (for example, a dashboard refresh).
- Edge and fog computing make hard real-time closed loops possible while still using cloud intelligence.

2.4 Architecture example

A bottling line: PLCs run the filling valves (device layer); an edge gateway checks fill level and sends commands if the level drifts (edge); a fog server coordinates the capper and labeller (fog); the cloud stores batch records and runs yield analysis every night (cloud).

Exam Q: Explain why a safety stop should not depend on a cloud connection.

3. Cloud-based Manufacturing Execution Systems (MES)

3.1 What is an MES?

A Manufacturing Execution System controls and tracks production on the shop floor. It sits between the planning system (ERP) and the machines (PLCs, SCADA).

- Functions: work order management, production tracking, resource allocation, quality management, material traceability, labour tracking, performance analysis and maintenance coordination.

3.2 Where MES fits in the automation pyramid

- Level 4: ERP (business planning, finance, orders).
- Level 3: MES (production execution and scheduling).
- Level 2: SCADA and HMI (supervision of processes).
- Level 1: PLCs and drives (control).
- Level 0: sensors and actuators (field devices).

3.3 Cloud-based MES

A cloud MES is delivered as a service (SaaS). Plants access it through web browsers and connect machines through an edge gateway.

- Advantages: quick deployment, automatic updates, multi-plant views, lower IT burden, pay-per-use pricing.
- Concerns: data security, dependence on the internet, integration with legacy systems, and the need for local fallback.
- Hybrid designs keep critical execution local and use the cloud for analytics and cross-site comparison.

3.4 Key KPIs an MES tracks

- OEE (availability x performance x quality).
- First-pass yield and scrap rate.
- Cycle time and throughput.
- Work-in-progress (WIP) levels.
- On-time delivery.

3.5 Example: traceability in pharmaceuticals

Each batch receives a unique ID. The MES records every operation, operator, machine setting, material lot and test result. If a problem is found later, the batch history can be traced in minutes instead of days.

KEY: MES connects "what should be made" (ERP) with "what is being made" (machines).

Exam Q: Name three functions of an MES and one benefit of moving it to the cloud.

4. Smart Sensors and RFID in Inventory Management

4.1 Smart sensors in inventory

Smart sensors on shelves, bins, pallets and vehicles track quantity, location, temperature, humidity and shock. They can alert staff when a bin is nearly empty or when cold-chain conditions are broken.

4.2 What is RFID?

Radio Frequency Identification (RFID) uses radio waves to read and write data on a tag attached to an object, without line of sight. A reader sends a signal; the tag responds with its unique ID.

4.3 Components of an RFID system

- Tag: a microchip with an antenna. Passive tags have no battery and draw power from the reader signal; active tags have their own battery and a longer range.
- Reader (interrogator): transmits and receives radio signals.
- Antenna: part of the reader that sends and receives.
- Middleware and database: filters the reads and stores them with time and location.

4.4 RFID frequency bands

- Low frequency (LF, about 125 to 134 kHz): short range, good near metal or liquids; used for animal tags and access control.

- High frequency (HF, 13.56 MHz): medium range; used for library books and payment cards (NFC is a form of HF RFID).
- Ultra high frequency (UHF, about 860 to 960 MHz): long range, fast bulk reading; used for pallets and supply chains.
- Microwave (2.45 GHz and above): very long range, used for toll and vehicle tracking.

4.5 Uses in inventory management

- Automatic stock counting without opening boxes.
- Goods receipt and dispatch with dock-door readers.
- Work-in-progress tracking on the shop floor (tagged pallets or carriers).
- Tool management: checking tools in and out of a crib.
- Reduced stock-outs and less manual error.

4.6 Advantages and limitations of RFID

- Advantages: no line of sight needed, many tags read at once, reusable tags, rich data.
- Limitations: metal and liquids can disturb signals; tag cost; read collisions when many tags are close; privacy concerns when tags are used on products sold to consumers.

KEY: RFID tells you WHICH item is where; barcodes need a clear line of sight and are read one at a time.

Exam Q: Compare a barcode and an RFID tag for tracking pallets in a warehouse.

5. Blockchain in Manufacturing

5.1 What is blockchain?

Blockchain is a distributed, append-only ledger. Records are grouped into blocks; each block contains a cryptographic hash of the previous block, so changing an old record breaks the chain. Many parties hold a copy and agree on new entries through a consensus mechanism.

5.2 Key ideas

- Distributed: no single owner controls the record.
- Immutable: past entries are very hard to change without detection.
- Transparent: all permitted parties see the same history.
- Consensus: entries are validated by agreed rules (for example, proof of authority in private networks).

5.3 Types of blockchain

- Public: open to anyone (for example Bitcoin, Ethereum).
- Private: run by one organisation.
- Consortium (permissioned): run by a group of known companies; common in supply chains.

5.4 Blockchain for secure supply chains

- Traceability: record each step from raw material to final product (origin, processing, shipping).
- Authenticity: prove that parts are genuine and not counterfeit.
- Compliance: store certificates, test results and audit trails that cannot be quietly edited.
- Trust between suppliers: parties who do not trust each other can share the same record.
- Payment and claims: automatic settlement when goods arrive in good condition.

5.5 Example

A food producer records harvest date, farm, temperature history from IoT sensors and certification on a consortium blockchain. Retailers scan a code and see the complete history. If a contamination is found, the affected batches can be identified within minutes.

5.6 Limitations

- Scalability and speed: public chains can be slow and costly.
- Garbage in, garbage out: blockchain cannot verify that the initial data (for example, a sensor reading) was true.
- Energy use in some consensus methods.

- Integration effort with existing ERP and MES.
- Governance: deciding who can write data and who can change rules.

Exam Q: Why does blockchain need a cryptographic hash link between blocks?

6. Smart Contracts and Decentralised Manufacturing Syst

6.1 Smart contracts

A smart contract is a program stored on a blockchain that runs automatically when agreed conditions are met. It removes the need for a human intermediary to enforce the agreement.

- Example: "When the supplier's delivery sensor confirms arrival of 500 kg of resin at the correct temperature, release the payment."
- Written in languages such as Solidity (Ethereum) or chaincode in Hyperledger Fabric.
- Once deployed, the rules are fixed; mistakes in code can be costly, so testing and audits are essential.

6.2 Smart contracts in manufacturing

- Automatic purchase order and payment on delivery.
- Quality-based payment (payment reduced if measured defect rate is high).
- Service-level penalties for late deliveries.
- Automatic reordering when inventory falls below a limit.

6.3 Decentralised manufacturing systems

In a centralised system, one controller or company decides all schedules. In a decentralised system, autonomous units (machines, cells, AGVs, even companies) negotiate and make local decisions based on rules and shared data.

- Advantages: resilience (failure of one unit does not stop the others), flexibility, faster local response, scalability.

- Challenges: achieving a good global result, coordination, trust and standards.

6.4 Example: distributed production scheduling

Each machine agent bids for jobs based on its load and capability. A job is assigned to the agent that offers the best cost and time. If one machine fails, its jobs are re-bid to other agents automatically.

6.5 Decentralised manufacturing in practice

- Distributed additive manufacturing: printing spare parts at many local sites from one verified digital file.
- Blockchain-based machine-to-machine payments for use of shared equipment.
- Supply networks in which suppliers and manufacturers share schedules through a common ledger.

KEY: Decentralised does not mean uncontrolled: rules, identity and standards still govern every participant.

Exam Q: Write one smart contract condition for a supply chain of your choice.

6.6 Blockchain consensus methods (quick notes)

- Proof of Work (PoW): miners solve hard puzzles; secure but energy-heavy; used by Bitcoin.
- Proof of Stake (PoS): validators are chosen according to the stake they lock; far less energy.
- Proof of Authority (PoA) and PBFT: a known set of trusted validators agrees on blocks; fast and suitable for private or consortium chains.

KEY: For factories, permissioned chains with PBFT-style or PoA consensus are usually the practical choice.

2.5 Edge hardware choices

- Programmable logic controllers with built-in communication (fast and certified for industry).
- Industrial PCs running Linux or Windows for data-heavy applications.
- Embedded boards such as single-board computers for simple sensor gateways.
- GPU or AI accelerator modules for running vision and anomaly models near the machine.
- Selection criteria: temperature range, vibration and dust rating, power supply, number of I/O ports, security features and long-term availability of parts.

1.6 Cloud security basics for manufacturing

- Encrypt data in transit (TLS) and at rest.
- Use strong identity management: unique accounts, multi-factor login, role-based access.
- Segment networks: keep the control network separate from the office network and expose only what is needed through a DMZ.
- Monitor logs continuously and keep backups in another region.
- Define who owns the data and where it may be stored (data governance).

KEY: Connecting machines to the cloud increases the attack surface; security must be designed in from the start.

3.6 Steps for rolling out a cloud MES

1. Map current processes: work orders, routing, quality checks and reporting.
2. Define KPIs and the business goal (for example, reduce WIP by 20 percent).
3. Select vendor and check integration with ERP, PLC, SCADA and quality systems.
4. Connect a pilot line through an edge gateway; test data accuracy against manual records.
5. Train operators and supervisors; design simple screens for the shop floor.
6. Go live on the pilot, measure KPIs for 4 to 8 weeks, then extend to other lines.
7. Plan for fallback: if the internet is down, the line should continue with local buffering.

Exam Q: Why is it important to test data accuracy during the pilot phase of an MES project?

5.7 Blockchain versus traditional database

Feature	Traditional database	Blockchain
Control	Single owner or administrator	Shared among participants
Edits	Records can be changed or deleted	Past blocks are protected by hashes
Trust	Trust in the administrator	Trust in the protocol and consensus
Speed	Very fast	Slower, depends on design
Best use	Internal records of one company	Shared records between many companies

- Blockchain is useful when parties do not fully trust one another and need a common record.
- If only one company needs the data, a well-secured database is usually simpler and cheaper.

7. Worked Examples and Revision

7.1 Worked example: RFID-based pallet tracking

- Tag each pallet with a UHF tag holding its unique ID and product code.
- Install readers at receiving, storage, production and dispatch doors.
- Each read is time-stamped and sent through the edge gateway to the WMS (warehouse management system).
- The system shows live stock levels and automatically flags a pallet that leaves the wrong dock.

7.2 Worked example: edge decision for a temperature alarm

- Cold-storage sensor reads 9 degrees Celsius; the limit is 8 degrees.
- The edge gateway triggers a local alarm and starts the backup cooler within 2 seconds.
- A summary is sent to the cloud for records and to the MES for the affected batch.
- Result: the batch is saved without waiting for an internet round trip.

7.3 Revision checklist

- Can you define IaaS, PaaS and SaaS with one example each?
- Can you explain why edge computing reduces latency?
- Can you draw the automation pyramid and place the MES?
- Can you describe the components of an RFID system?
- Can you compare LF, HF and UHF RFID?
- Can you explain blockchain immutability in simple words?
- Can you describe a smart contract with an example?

7.4 Short answer practice

Exam Q: What is the difference between fog and edge computing?

Exam Q: Name two KPIs that an MES tracks and explain one.

Exam Q: Why is a consortium blockchain often preferred over a public one in supply chains?